

Cloud Computing

MODULE-1

Ahmed

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Q.1) Define cloud computing & identify the main features of Cloud computing.

- cloud computing is a technology in which computing services like storage, servers, software & networks are provided over the internet.
- Users can use resources whenever needed and pay only for what they use, without owning physical hardware.

— Main Features of Cloud Computing:

- On-demand Service — Resources are available anytime.
- Pay as you use — Pay only for used services.
- Resource Pooling — Resources are shared among many users.
- Scalability — Resource can increase or decrease easily
- Cost Effective — No need to buy expensive hardware

Q.2) Define virtualization. What are the advantages of virtualization?

- Virtualization is a technology that allows one physical computer to be divided into multiple virtual machines using software called a hypervisor.
- Each virtual machine works like an independent computer with its own operating system.

→ Advantages of Virtualization

- Better use of hardware — one system runs many machines
- Cost saving — Less physical hardware required
- Easy scalability — Virtual machines can be added or removed easily
- Easy backup & recovery — Virtual machines can be copied or restored easily.
- Supports cloud computing — ~~Foundation technology of cloud computing~~

Q.3) What are the components of a distributed system? Explain.

→ A distributed system consists of multiple computers that work together & appear as a single system to the user.

— Components of Distributed System:—

(I) Client:

- Requests services from the system.
- Example, User computer, mobile device.

(II) Server:

- Provides services to clients.
- Stores data & performs processing.

(III) Network:

- Connects clients & servers.
- Enables communication between systems.

(IV) Middleware:

- Act as a bridge between client & server.
- Handles communication, data transfer & coordination.

(V) Communication Protocols:

- Rules for data exchange.
- Example, TCP/IP, HTTP.

Q4) What are the elements of distributed computing? Explain.

→ Distributed computing consists of multiple computers working together to perform a common task efficiently.

— Elements of Distributed Computing

(i) Multiple Nodes (computers)

- Many computers work together
- Each node performs a part of the task.

(ii) Communication Network

- connects all nodes
- Enables data & message transfer

(iii) Concurrency

- Multiple tasks run at the same time.
- Improves system performance.

(iv) Scalability

- Easy to add or remove nodes
- Supports system growth.

(v) Transparency

- User sees the system as a single computer
- Hides complexity from users.

Q5) Describe Classification or taxonomy of virtualization at different levels.

→ Virtualization can be classified based on the level at which it is implemented.

The main types are as follows:

(I) Hardware Level Virtualization:-

- Virtualization is done at hardware level.
- Uses hypervisor software
- Multiple virtual machines run on one physical system.
- Each VM has its own operating system.
- Eg: VMware, Hyper-V

(II) Operating System Level Virtualization:-

- Multiple virtual environments share the same OS kernel
- No separate OS for each virtual machine.
- Faster & uses fewer resources.
- Eg - Containers

(III) Server Virtualization:-

- A single physical server is divided into multiple virtual servers.
- Improves server utilization.
- Commonly used in data centers.

Q.6) Differentiate between Parallel & Distributed Computing.

→ Parallel Computing

- Uses a single system
- Processors are tightly coupled
- Uses shared memory
- Faster execution for one task
- Less fault tolerant
- Mostly used in supercomputers
- Limited geographical area

Distributed Computing

- Uses multiple systems
- Systems are loosely coupled
- Each system has separate memory
- Better scalability
- Highly fault tolerant
- Used in networks & cloud
- Can be geographically distributed

Q.7) What is Web Services? Explain the different types of Web Services.

- A web service is a software system that allows different applications to communicate with each other over the internet using standard protocols.
- It helps share data & functions between programs, even if they are developed on different platforms or languages.

— Types of Web Services: —

(i) SOAP web Services:

- SOAP stands for Simple Object Access Protocol
- Uses XML for message exchange
- More secure & reliable
- Used mainly in enterprise application.

(ii) REST web Services:

- REST stands for Representational State Transfer
- Uses HTTP protocol
- Simple & lightweight
- Faster than SOAP
- Widely used in cloud & mobile application.

Q.8) What are the technologies that are supported by Cloud computing?

→ cloud computing is supported by several technologies that help in efficient resource sharing, scalability & service delivery.

— Technologies Supporting Cloud Computing:—

- (i) Virtualization: Allows multiple virtual machines on one physical system
- (ii) Distributed Computing: Multiple computers works together as one system
- (iii) Web Services: Enables communication between application over the ^{internet}
- (iv) Utility Computing: Resources provided on pay-as-you-use basis
- (v) Automation: Automatically manages resources & workloads.

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